

TOTHBOT — KRAKEN SYSTEM ARCHITECTURE OVERVIEW

The Complete System — All Architectural Decisions + All Flow Diagrams

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PURPOSE: This document captures all architectural decisions, FP/DP analyses, and system specifications from the TothBot Kraken architecture sessions of March 2026. Includes all six system flow diagrams showing connections between Kraken, TothBot, and CIATS as one unified organism.

Diagrams in this document (dv1_12 — diagrams unchanged from dv1_10):

Diagram 1 — Kraken API Connection Architecture

(dv1_6: amend_order primary, batch_cancel added, WS ping A-7, zombie A-8, balances seq A-9, per-pair status, L3 URL corrected)

(dv1_7: executions seq gap detection A-10, GetOpenOrders gap recovery)

(dv1_9: GAP-A order_status:true CRITICAL FIX, GAP-C price_increment, GAP-D qty_min/cost_min, GAP-E max_size=10MB, IMP-11 maxratecount, IMP-13 warnings[])

(dv1_10: GAP-B stp_type underscore CORRECTION, A-21 deadline parameter, IMP-16 aiohttp ClientTimeout, IMP-17 TCPConnector, IMP-18 ticker bbo)

(dv1_12: text-only updates — diagrams unchanged)

Diagram 2 — TothBot Trading Pipeline

(dv1_6: per-pair instrument status check added before Gate 1)

Diagram 3 — Exit Architecture: Ground Zero Redesign (dv1_6: title updated)

Diagram 4 — Regime Taxonomy: 6-Regime Grid + Two-Level Tagging (dv1_6: title updated)

Diagram 5 — CIATS Architecture: Data Flows and Improvement Cycle (dv1_6: title updated)

Diagram 6 — Complete System: One CIATS-Owned Organism (dv1_9: GAP-A through F, IMP-11 through 17) (dv1_10: GAP-B stp_type correction, A-21 deadline, IMP-16/17 aiohttp, IMP-18 bbo, IMP-19 regime stagger)

1. System Identity and Mission

1.1 What TothBot Is

TothBot is an automated cryptocurrency spot trading system on Kraken exchange. It runs 24/7/365 on a Hetzner VPS, processes 5-minute OHLC candle close events as its system clock, and executes long-only trades using a momentum-based signal engine.

TothBot is not an external system connected to Kraken — it IS Kraken's event processor. Every action is a response to a Kraken push event. No timers. No polling. Idle and silent until Kraken speaks.

1.2 The One Organism Principle

Three functional zones, one organism:

Kraken: generates events, executes orders, holds resting TP limit orders and off-book emergency SL stop orders on matching engine

TothBot: processes Kraken events, executes trading strategy, manages positions

CIATS: learns from event records, continuously improves processing rules

PRIMARY MISSION: Maximize profit. Minimize loss. With an emphasis on minimizing loss.

Loss prevention takes priority over profit generation. Everything else is secondary.

1.3 Operating Environment

Exchange: Kraken Spot (Long Module – Short deferred)

Language: Python 3.12.3

VPS: Hetzner CPX22 — 87.99.141.44 — Ashburn VA

(Kraken Spot servers: AWS us-west-2 Oregon)

★ Evaluate Hillsboro OR [dv1_6 IMP-01 — adjacent to Kraken AWS]

System Clock: Kraken OHLC 5-minute candle close event (WS v2 push)

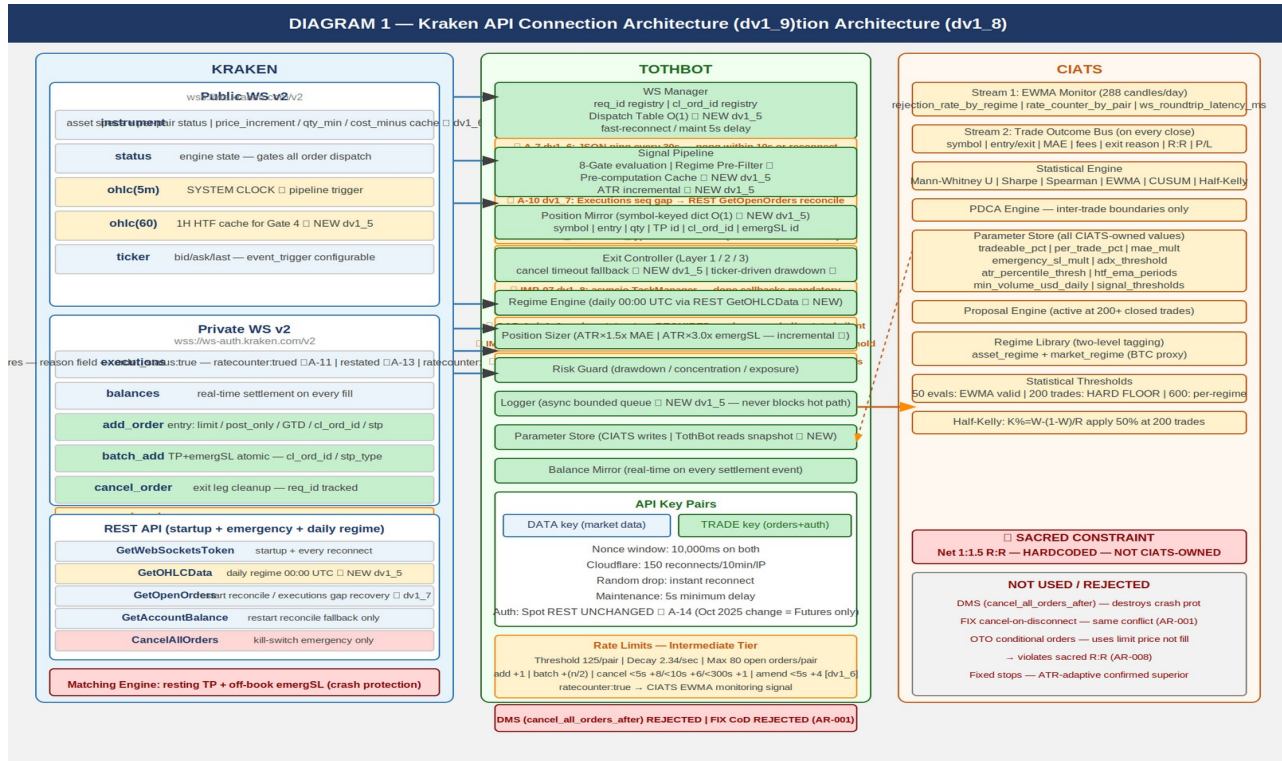
Fee Baseline: Maker 0.16% / Taker 0.26% (CIATS monitors actuals)

Operation: 24/7/365 – CIATS may add stand-down periods via PDCA if data supports

Starting Portfolio: \$1,000 Long Module

Sacred constraint: Net 1:1.5 R:R — HARDCODED — the only number not owned by CIATS

2. Kraken API Connection Architecture



All connection points verified across 3 independent analysis passes (15 sources) plus 2 deep validation sessions (144+ sources total) plus 10-search FP/DP Kraken validation [dv1_6] (126 sources).

2.1 Key Connection Points

Public WS v2:

- instrument — asset specs + per-pair status cache ★ dv1_6
- price_increment / qty_min / cost_min ★ GAP-C/D dv1_9
- status — global engine state (online/post_only/cancel_only/maintenance)
- ohlc(5m) — SYSTEM CLOCK (pipeline trigger, fires on trade events)
- ohlc(60) — 1H HTF cache for Gate 4 ★ NEW dv1_5
- ticker — price data, event_trigger:bbo (pairs with open positions) or event_trigger:trades (pairs without). WS Manager manages per-pair.

Private WS v2 inbound:

- executions — fills/expiration/SL-TP triggers, ratecounter:true (A-1), snap_orders:true, snap_trades:true, order_status:true on reconnect (A-3), sequence gap detection ★ A-10 dv1_7
- CRITICAL ★ GAP-A dv1_9: order_status:true MANDATORY — without it amended and restated exec_types are silently dropped, breaking A-12 and A-13
- balances — real-time settlement, sequence gap detection ★ A-9 dv1_6

Private WS v2 outbound:

- add_order - entry:
- limit/post_only/GTD/cl_ord_id/stp_type:cancel_newest/deadline:now+5s
- amend_order - PRIMARY amendment method (edit_order DEPRECATED ★ dv1_6)
- batch_add - SL+TP atomic on fill. cl_ord_id/stp_type:cancel_newest/deadline:now+5s
- cancel_order - exit leg cleanup, req_id tracked
- batch_cancel - Full Halt multi-order cancel ★ dv1_6

All messages carry req_id (A-2), cl_ord_id (A-2), stp_type:cancel_newest (A-4) [WS v2 underscore]

REST:

- GetWebSocketsToken - startup/reconnect
- GetOHLCData - daily regime computation 00:00 UTC ★ NEW dv1_5

CRITICAL: exclude response[-1] always ★ dv1_6

- GetOpenOrders - restart reconcile fallback / executions gap recovery ★ dv1_7
- GetAccountBalance - restart reconcile fallback / balances gap recovery ★ dv1_6
- CancelAllOrders - kill switch emergency only
- Status API - startup proactive maintenance check (AR-038 dv1_12)

GET <https://status.kraken.com/api/v2/>

[scheduled-maintenances/upcoming.json](#)

No authentication. Informational only. Does not block startup.

Matching Engine:

Resting TP limit orders and off-book emergency SL stop orders survive any TothBot failure independently — crash protection always active

EVALUATED AND REJECTED:

- cancel_all_orders_after (WS v2 DMS) - destroys crash protection (AR-001)

FIX API cancel-on-disconnect — same architectural conflict (AR-001)

OTO conditional orders — uses limit price not fill price, violates R:R (AR-008)

2.2 Four Kraken Engine States

- online - All orders. Signal pipeline runs normally.
- post_only - TothBot limit+post_only entries ARE valid. Trading continues.
- cancel_only - No new entries. Exit Controller can cancel. Resting orders active.
- maintenance - No orders, no cancels. Resting TP and emergSL active on engine.

2.3 API Rate Limits — Intermediate Tier (C-2)

Rate limits shared across REST, WebSockets, and FIX — no separate per-protocol budget.

Rate counter threshold (per pair): 125

Rate counter decay (per second): 2.34

Max open orders (per pair): 80

Add order fixed cost: +1

Amend order cost ★ dv1_6 (PRIMARY): +1 fixed, +3(<5s), +2(<10s), +1(<15s)

Edit order (DEPRECATED ★ dv1_6): +1 fixed, +6(<5s), +5(<10s) — DO NOT USE

Batch add fixed cost: +(n/2)

Cancel order cost (< 5s old): +8

Cancel order cost (< 10s old): +6
Cancel order cost (< 300s old): +1
Batch cancel cost ★ dv1_6: Cumulative sum of individual order cancel costs

amend_order is 43% cheaper than edit_order for orders resting <5s (+4 vs +7).
amend_order preserves queue position (unlike cancel+resubmit).

2.4 WS Operational Requirements (A-1 through A-10)

A-1 Rate counter monitoring via executions channel:
executions subscription accepts ratecounter:true. Real-time rate counter per pair included at zero additional connection cost. CIATS EWMA Monitor adds rate_counter_by_pair as third monitoring signal. Assign to 0511003 and 0211001.

A-2 req_id and cl_ord_id on all outbound WS messages:
All outbound messages carry req_id (client-specified integer). All add_order and batch_add additionally carry cl_ord_id (up to 18 characters). TothBot assigns cl_ord_id at dispatch time before Kraken response. Position Mirror records cl_ord_id immediately at dispatch — no async gap. WS Manager maintains pending-request registry. Assign to 0511001, 0511003, 0511005.

A-3 Round-trip latency monitoring and WS reconciliation:
All WS v2 responses include time_in and time_out to nanosecond precision. Logger records these fields. CIATS EWMA Monitor adds ws_roundtrip_latency_ms. executions channel supports snap_orders:true on subscription — preferred over REST GetOpenOrders for restart reconciliation. Assign to 0511001, 0511006, 0211001.

A-4 stp_type:cancel_newest on all WS v2 add_order and batch_add calls:
All add_order and batch_add include stp_type:cancel_newest [WS v2 underscore format]. REST API uses cancel-newest [hyphen] — different protocols, different formats (A-20). Self Trade Prevention prevents TothBot from inadvertently filling against its own resting orders.

A-5 Reconnect behavior distinction:
(1) Random drop: reconnect instantly, up to a handful of attempts.
(2) After confirmed trading engine maintenance: minimum 5-second delay.
Both implemented as separate paths in WS Manager. Assign to 0511001.

A-6 (dv1_5) OTO conditional orders – REJECTED:
OTO uses limit price not actual fill price — violates sacred R:R. Creates Position Mirror race condition. batch_add on fill confirmed correct. See AR-008.

A-7 (dv1_6) ★ WS Ping / Keepalive Strategy:
Kraken WS server disconnects after ~60 seconds of inactivity. Cloudflare idle timeout ~100 seconds. WS Manager MUST send {"method":"ping"} on each active connection every 30 seconds. Expects {"method":"pong"} within 10 seconds. No pong within 10 seconds = dead connection. Reconnect immediately. Application-level ping, distinct from TCP keepalive. Assign to 0511001, 1011002.

A-8 (dv1_6) ★ Zombie Connection Detection:

A WS connection can appear alive (ping/pong passing) while delivering no real market data. WS Manager MUST track `last_real_data_time` per connection using `time.monotonic()`. Only actual market data events reset this timestamp — heartbeat messages alone do NOT. If elapsed > 90 seconds: log `ZOMBIE_CONNECTION_DETECTED`, alert operator, reconnect. Assign to 0511001, 1011002.

A-9 (dv1_6) ★ Balances Channel Sequence Gap Detection:

Every balances channel message carries a sequence integer. WS Manager MUST track last-seen sequence. Any gap (`skip > 1`) indicates a missed ledger event. On gap: alert immediately, trigger REST `GetAccountBalance` reconciliation. Missed balance events cause incorrect available capital calculation — direct loss risk. Assign to 0511001, 1011002.

A-10 (dv1_7) ★ Executions Channel Sequence Gap Detection:

Every executions channel message carries a sequence integer. WS Manager MUST track last-seen sequence independently of balances. Any gap (`skip > 1`) indicates missed order/fill events. On gap: alert immediately, trigger REST `GetOpenOrders` reconciliation to restore Position Mirror accuracy. Missed execution events mean TothBot does not know a position is open or closed — direct loss risk. Higher severity than A-9 (balances): a missed fill leaves Position Mirror in stale state; the next pipeline evaluation operates on incorrect position data. Sequence tracking resets on each new subscription (reconnect). Do not gap-detect across sessions. Assign to 0511001, 1011002.

A-11 (dv1_8) ★ Executions Channel: `cancel_reason` Deprecated – Use `reason` Field:

The `cancel_reason` field in executions channel events is DEPRECATED since WS v2.0.0 (December 2023). All code parsing cancellation events MUST read `msg["reason"]`, not `msg["cancel_reason"]`. `cancel_reason` returns `None` on all modern messages. Silent reason loss corrupts CIATS EWMA `rejection_rate_by_regime` monitoring accuracy. Assign to 1011002 WM-EXE-002 through WM-EXE-004.

A-12 (dv1_8) ★ Executions Channel: Complete `exec_type` Enum Handling Required:

WS Manager MUST handle all `exec_type` values: `pending_new`, `new`, `trade`, `filled`, `iceberg_refill`, `canceled`, `expired`, `amended`, `restated`, `status`. Using a partial dispatch table causes unknown `exec_type` events to silently fall through. Use `handle_exec_unknown` for unrecognized values — log full payload, never silently ignore. TothBot primary events: `trade` (trigger `batch_add`), `filled`, `expired` (log GTD expiry), `canceled` (log with reason). `amended` and `restated`: log only, no Position Mirror change. Assign to 1011002 WM-EXE-005 through WM-EXE-007.

A-13 (dv1_8) ★ Executions Channel: `restated exec_type` Must Not Trigger Position Change:

`restated` fires when Kraken's matching engine amends an order for maintenance purposes. It is NOT a fill, NOT a cancel, NOT a user action. On `restated`: (1) log with reason field, (2) alert operator (informational), (3) do NOT update Position Mirror, (4) if the `restated` order is a resting TP or `emergSL`, escalate to HIGH severity alert and

queue a reconciliation check on the next pipeline cycle. The restated event on a resting crash-protection order indicates Kraken modified our stop — must be visible. Assign to 1011002 WM-EXE-008 through WM-EXE-010.

A-14 (dv1_8) ★ Spot REST Authentication: Confirmed Unchanged — PL-014 RESOLVED: The October 1, 2025 authentication breaking change applied to Futures REST ONLY. Kraken Spot REST authentication is UNCHANGED: HMAC-SHA512 of (URI path + SHA256 (nonce + POST data)) using base64-decoded secret. No code changes required. PL-014 is fully resolved. No Bill verification action required. Assign to 0411001 WS-AUTH-001 through WS-AUTH-003.

A-15 (dv1_9) ★ Executions Channel: order_status:true MANDATORY — GAP-A CRITICAL FIX: Without order_status:true in the executions subscription, the channel only delivers: new, filled, canceled, expired. The exec_types amended and restated are SILENTLY DROPPED. This means: (1) A-13 restated protection for crash-protection orders is completely non-functional without this flag — the restated event never arrives. (2) A-12 dispatch table handles 10 exec_types but receives only 4. The subscription is already specified correctly (WS-EXE-001 in 0411001) but this architectural consequence must be explicitly documented. Assign to 0411001 WS-OSTA-001 through WS-OSTA-003, 1011002 WM-EXE-011/-012.

A-16 (dv1_9) ★ OHLC Channel: interval_begin Replaces Deprecated timestamp — GAP-B: The Kraken WS v2 OHLC channel deprecated the timestamp field. Subscription ACK warns: "timestamp is deprecated, use interval_begin". TothBot MUST parse interval_begin (RFC3339 nanosecond string) as the candle start time, not timestamp. Assign to 0411001 WS-OHLC-002 through -005, 1011002 WM-OHLC-001 through -003.

A-17 (dv1_9) ★ Instrument Channel: price_increment / qty_min / cost_min — GAP-C/D: (GAP-C) The instrument channel deprecated tick_size. Use price_increment exclusively. Subscription ACK warns: "tick_size is deprecated, use price_increment". (GAP-D) The instrument channel provides qty_min (minimum order quantity) and cost_min (minimum order cost = price × qty). Gate 8 Position Sizer MUST validate both before submitting any add_order. Violation causes immediate order rejection. Assign to 0411001 WS-INST-001 through -007, 1011002 WM-INST-001 through -005.

A-18 (dv1_9) ★ WebSocket Library max_size = 10MB — GAP-E: Python websockets library default max_size = 1MB. The Kraken instrument channel snapshot for 500+ pairs can exceed 1MB, causing PayloadTooBig and silent disconnect. All WS connect() calls MUST set max_size=10*1024*1024 (10MB) and open_timeout=10. Assign to 0411001 WS-LIB-001 through -003, 1011002 WM-LIB-001 through -003.

A-19 (dv1_9) ★ maxratecount from Executions ACK is Operative Rate Threshold — IMP-11: The executions subscription ACK returns maxratecount — the actual operative rate counter ceiling for this user tier. TothBot MUST extract and use maxratecount as the operative threshold for all rate counter monitoring, not the hardcoded value 125. If tier changes, maxratecount automatically updates on next reconnect. Assign to 0411001 WS-RL-003, WS-EXEC-001a, 1011002 WM-RATE-001 through -003.

A-20 (dv1_10) ★ stp_type Format CORRECTION – WS v2 Uses Underscore, Not Hyphen:
Official Kraken WS v2 documentation and batch_add examples use underscore format:
cancel_newest, cancel_oldest, cancel_both
REST API uses hyphen format: cancel-newest, cancel-oldest, cancel-both.
These are DIFFERENT protocols with DIFFERENT formats. All WS v2 outbound messages
(add_order, batch_add) MUST use the underscore format. Using the hyphen format in
WS v2 will cause order rejection or silent failure. This was an error in all
prior dv1_9 and earlier specifications. Corrected here as AR-032.
Assign to 0411001 WS-STP-001 through -003, 1011002 WM-STP-001 through -003.

A-21 (dv1_10) ★ deadline Parameter on All WS v2 Order Submissions:
Both add_order and batch_add support a deadline parameter: an ISO 8601 UTC
timestamp string. The engine rejects any order arriving after this time.
Valid range: 500ms to 60 seconds from current time. Default: 5 seconds.
TothBot standard: deadline = current UTC time + 5 seconds.
Provides protection against stale order placement due to network delay or
event loop pauses. Without deadline, a delayed add_order could place an entry
at a price that has already moved adversely.
CRITICAL: deadline also applies to batch_add. A delayed batch_add with incorrect
TP/emergSL prices due to fill price slippage is prevented by deadline enforcement.
Generate: deadline = (datetime.utcnow() + timedelta(seconds=5)).strftime(
'%Y-%m-%dT%H:%M:%S.%f')[:-3] + 'Z'
Assign to 0411001 WS-DL-001 through -004, 1011002 WM-DL-001 through -004.

A-22 (dv1_10) ★ aiohttp ClientSession Timeout and Connector Configuration:
All REST calls use a single shared aiohttp.ClientSession with explicit timeout and
connector configuration. Without explicit timeout, REST calls can hang indefinitely
if Kraken REST servers become unresponsive. This blocks the asyncio event loop
for aiohttp operations and starves other tasks.
Required configuration:
timeout = aiohttp.ClientTimeout(total=10, connect=5, sock_read=8)
connector = aiohttp.TCPConnector(limit=10, limit_per_host=5, force_close=False)
session = aiohttp.ClientSession(timeout=timeout, connector=connector)
total=10: maximum 10 seconds for complete request (connect + send + read).
connect=5: maximum 5 seconds to establish TCP connection.
sock_read=8: maximum 8 seconds waiting for server response after sending.
force_close=False: enables HTTP keepalive — connection reused across calls.
Assign to 1011002 WM-REST-006 through -011, 1011001 BP-HTTP-001 through -004.

A-23 (dv1_10) ★ Ticker event_trigger per Position State:
The ticker channel event_trigger parameter controls when updates are published.
event_trigger:bbo — fires on every change to best bid or ask price.
event_trigger:trades — fires only when a trade occurs (matching engine event).
For pairs WITH open positions: use event_trigger:bbo.
Rationale: For long positions, the bid price is the realizable exit price.
bbo fires immediately when the bid drops, giving the Exit Controller maximum
reaction time to detect MAE threshold breach. trades-only can miss bid drops
that occur without a matching trade.

For pairs WITHOUT open positions: use event_trigger:trades.

Rationale: Reduces unnecessary event volume. Signal pipeline uses last price not bid, so trades trigger is sufficient for scoring.

WS Manager adjusts per-pair event_trigger when Position Mirror changes state.

Implementation: re-subscribe ticker for that pair with updated event_trigger.
Assign to 0411001 WS-TKR-001 through -004, 1011002 WM-TKR-001 through -004.

A-24 (dv1_10) ★ Regime Engine GetOHLCData Rate Limit Staggering:
REST public GetOHLCData is rate-limited by IP address + currency pair at 1/second.
Calling it for all monitored pairs simultaneously at 00:00 UTC triggers rate limits.
All GetOHLCData calls in the Regime Engine MUST be staggered with asyncio.sleep(1.1) between each pair to stay below the rate limit ceiling.
Example for 50 pairs: $50 \times 1.1s = 55s$ total. Completes well before first 5m candle of the new day (~05:00 UTC for the first significant candle after the daily open).
On rate limit error (HTTP 429): exponential backoff, log REGIME_OHLC_RATE_LIMITED, retry up to 3 times. If all fail: use stale regime (per RE-SCH-004).
Assign to 1011012 RE-RL-001 through -004.

2.5 Data Architecture: Pre-Computation Cache (NEW dv1_5 — I-2)

HOT PATH RULE: The 5-minute pipeline hot path reads cached values ONLY. No computation occurs during pipeline evaluation. All indicators maintained as incremental running calculations updated on each event trigger.

IMP-06 (dv1_8) ★ aiohttp Mandatory for All REST Calls:
All Kraken REST calls MUST use aiohttp (async HTTP). Never use the requests library inside asyncio — it blocks the event loop during the entire call duration. Create one aiohttp.ClientSession at TothBot startup, reuse for all REST calls, close at shutdown.
Assign to 1011002 WM-REST-001 through WM-REST-005.

IMP-07 (dv1_8) ★ asyncio TaskManager Pattern — Mandatory for 24/7 Operation:
All asyncio.create_task() calls MUST store a strong reference AND register a done callback via task.add_done_callback(task_set.discard). Failure to do so causes memory leaks in long-running processes: the task set grows indefinitely until OOM. All task creation routed through a central create_task() helper function. Long-running tasks (ping, zombie, watchdog) stored as named instance variables, cancelled and awaited explicitly during reconnect cleanup.
Assign to 1011002 WM-TASK-001 through WM-TASK-004.

IMP-08 (dv1_8) ★ systemd Watchdog Integration — Frozen Process Detection:
TothBot MUST run as a systemd service with WatchdogSec=120 and Type=notify. A separate asyncio.Task pings systemd sd_notify every 30 seconds via WATCHDOG=1 message. If the event loop hangs (frozen process), the watchdog stops, systemd detects no ping within 120s, and restarts TothBot. Without this, a frozen-but-alive TothBot with open positions is undetected indefinitely.
Assign to 1011013 VPS_Deployment_Coding_Spec VD-WD-001 through VD-WD-006.

IMP-16 (dv1_10) ★ aiohttp ClientSession Timeout – Required for REST Reliability:
 All REST calls via aiohttp MUST use explicit timeout and connector configuration.
 Without explicit timeout, a degraded Kraken REST server causes asyncio tasks to
 hang indefinitely, blocking reconnect/gap recovery sequences.
 ClientTimeout(total=10, connect=5, sock_read=8) applied at session creation.
 See A-22 for full specification. Assign to 1011002 WM-REST-006 through -011.

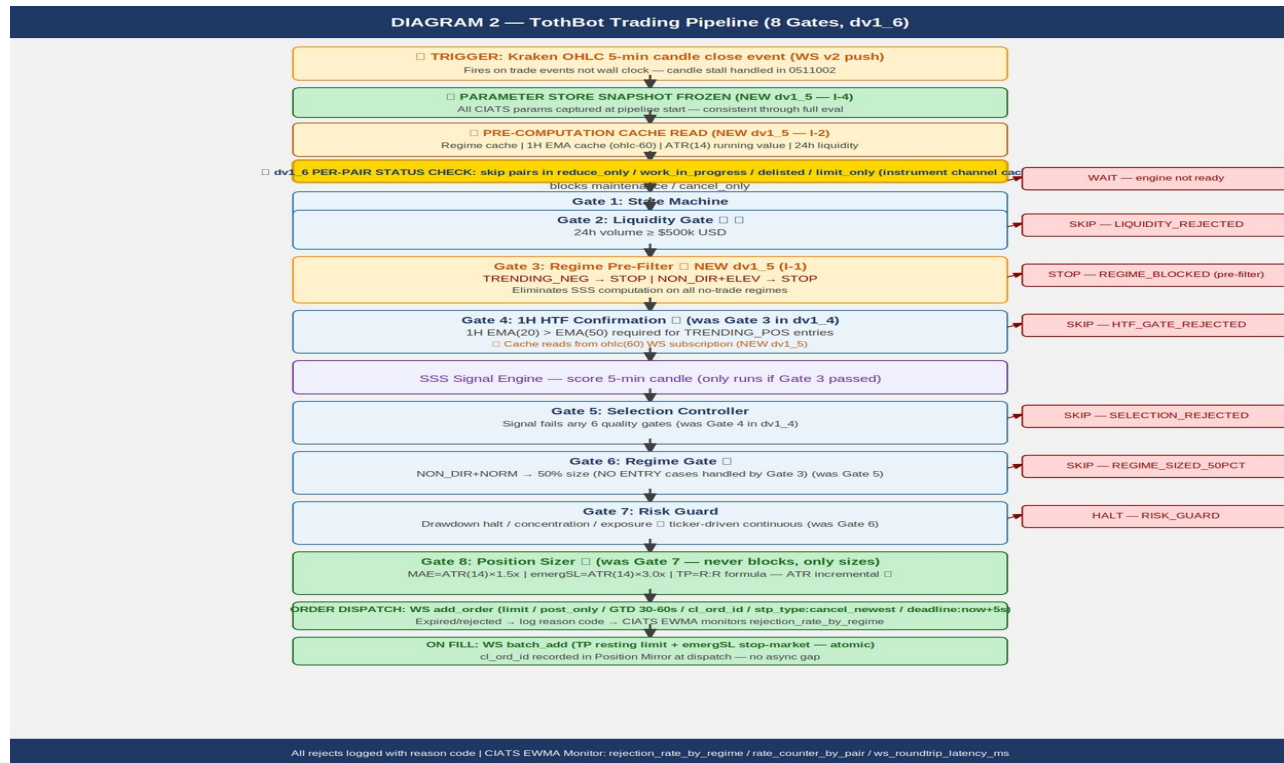
IMP-17 (dv1_10) ★ aiohttp TCPConnector for Connection Reuse:
 TCPConnector(limit=10, limit_per_host=5, force_close=False) enables HTTP keepalive.
 Without this, aiohttp closes the TCP connection after every REST call and re-opens
 it on the next call. For GetWebSocketsToken (called every reconnect) and
 GetOpenOrders (called on gap detection), connection reuse materially reduces
 latency.
 See A-22 for full specification. Assign to 1011002 WM-REST-009 through -011.

IMP-18 (dv1_10) ★ Ticker event_trigger:bbo for Open Position MAE Detection:
 Pairs with open positions MUST use event_trigger:bbo on ticker subscription.
 bbo fires immediately on any best bid/ask price change — not just on trades.
 For long positions, bid = realizable exit price. bbo maximizes Layer 2 exit
 detection speed. Pairs without open positions use event_trigger:trades.
 See A-23 for full specification. Assign to 0411001, 1011002.

IMP-19 (dv1_10) ★ Regime Engine REST Rate Limit Stagger Required:
 All GetOHLCData calls in the Regime Engine must use asyncio.sleep(1.1) between
 pairs to stay within public REST rate limits. Without stagger, simultaneous calls
 for 50+ pairs at 00:00 UTC hit the rate limit and corrupt regime classification
 for all affected pairs. See A-24. Assign to 1011012 RE-RL-001 through -004.

Value	Pre-computed by	Update Trigger	Cache Read On
Regime classification	Regime Engine	Daily 00:00 UTC REST	GetOHLCData Every 5m eval (Gate 3+6)
[CRITICAL: exclude response[-1] ★]			
1H EMA(20) and EMA(50)	Signal Pipeline	1H candle close ohlc(60)	WS event Gate 4
(parse interval_begin ★ A-16 dv1_9)			
ATR(14) per pair	Signal Pipeline	Each 5m candle close (incremental)	Gate 8 + Exit Controller
24h liquidity per pair (USD)	Signal Pipeline	Daily ticker or daily OHLC pull	Gate 2
Per-pair instrument status ★	WS Manager	instrument channel events (live)	Gates 1/2 pre-filter

3. TothBot Trading Pipeline



The pipeline fires on every 5-minute OHLC candle close event. Synchronous and event-driven.
8 gates. Gate 3 (Regime Pre-Filter) new in dv1_5. Per-pair status check new in dv1_6.

NOTE — OHLC fires on trade events, not wall clock: If no trades occur after an interval border, no message sent until next trade. Liquidity Gate (\$500k/day) reduces this risk. Same stall behavior applies to ohlc(60). TothBot must handle stalled system clock. Addressed in 0511002 Signal Pipeline Specification.

3.1 Gate Summary

PRE-STEP: Parameter Store Snapshot Frozen (NEW dv1_5 — I-4)
CIATS params captured as read-only snapshot at pipeline start. Consistent throughout full eval. CIATS writes applied only at inter-trade boundaries.

PRE-STEP: Pre-Computation Cache Read (NEW dv1_5 — I-2)
Hot path reads: regime cache | 1H EMA cache (ohlc-60) | ATR(14) running value | 24h liquidity | per-pair status. Zero computation on hot path.

★ dv1_6: Per-pair Instrument Status Check
Skip pairs in `reduce_only`, `work_in_progress`, `delisted`, `limit_only`, `cancel_only`, or maintenance state ★ dv1_12.
Only online or `post_only` pairs proceed into pipeline.
Prevents wasted rate counter. Full status list per 0411001 WS-INST-008.

Gate 1: State Machine

Kraken engine = maintenance or cancel_only: WAIT

Gate 2: Liquidity Gate ★

24h volume < \$500k USD: SKIP → LIQUIDITY_REJECTED

Gate 3: Regime Pre-Filter ★ NEW dv1_5 (I-1)

TRENDING_NEG (any vol) or NON_DIR+ELEVATED: STOP → REGIME_BLOCKED

Eliminates SSS computation on all no-trade regimes.

Gate 4: 1H HTF Confirmation ★ (was Gate 3)

1H EMA(20) <= EMA(50) for TRENDING_POS entries: SKIP → HTF_GATE_REJECTED

Cache read from ohlc(60) WS subscription (NEW dv1_5).

SSS Signal Engine

Scores 5-min candle. Only reached if Gate 3 pre-filter passed.

Gate 5: Selection Controller (was Gate 4)

Signal fails any of 6 quality gates: SKIP → SELECTION_REJECTED

Gate 6: Regime Gate ★ (was Gate 5)

NON_DIR+NORMAL: 50% position size. All NO ENTRY cases blocked by Gate 3.

Gate 7: Risk Guard (was Gate 6)

Drawdown halt / concentration / exposure. Pass or HALT.

★ Ticker-driven continuous drawdown monitoring between candle closes (I-7 dv1_5)

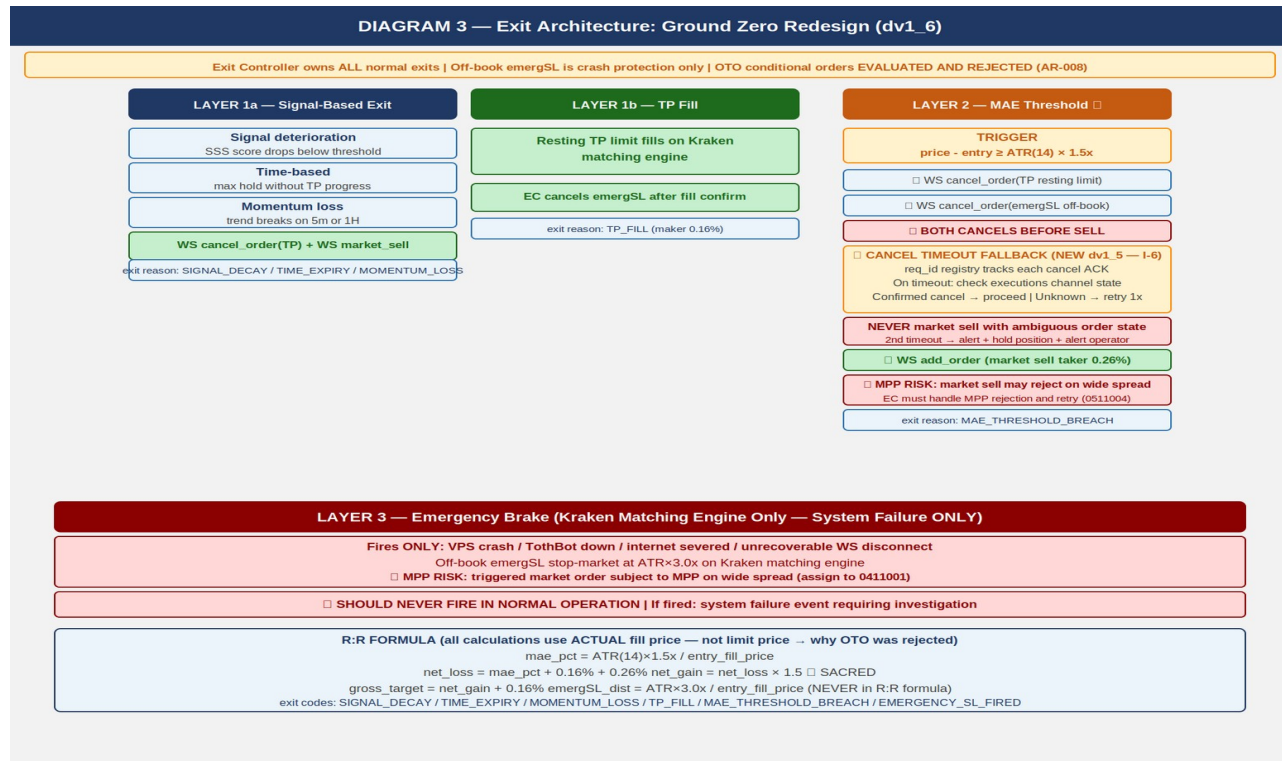
Gate 8: Position Sizer ★ (was Gate 7)

MAE = ATR(14) x 1.5x (incremental), emergSL = ATR(14) x 3.0x, TP = R:R formula.

Never blocks — only sizes.

EVIDENCE BASE: 15-20% higher win rates with daily+hourly confirmation (5 independent sources 2024-2025). Market Cipher: accuracy drops 65% in TRENDING to 45-50% in NON_DIRECTIONAL. Break-even 40% for 1:1.5 R:R.

4. Exit Architecture — Ground Zero Redesign for Kraken



Rebuilt from zero for Kraken. Exit Controller owns all normal exits. Kraken off-book emergency SL stop order is crash protection only. OTO evaluated and rejected (AR-008).

4.1 Three-Layer Summary

Layer 1a — Signal-Based Exit:

Signal decay / time expiry / momentum loss. WS cancel TP + market sell.

Layer 1b — TP Fill:

Resting TP limit on Kraken matching engine fills. EC cancels emergSL after fill confirm.

Layer 2 ★ — MAE Threshold:

price - entry $\geq$ ATR(14) $\times$ 1.5x.

SEQUENCE CRITICAL: (1) cancel TP (2) cancel emergSL THEN (3) market sell.

CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6):

req_id registry tracks each cancel ACK. On timeout: check executions channel state.

Confirmed cancelled: proceed. State unknown: retry once. Second timeout: ALERT + hold position + alert operator. NEVER issue market sell with ambiguous order state.

Assign to 0511001 and 0511004.

MPP RISK (C-1 dv1_4): market sell may reject on wide spread. EC MUST handle MPP rejection and retry. Assign to 0511004.

Layer 3 ★ — Emergency Brake:

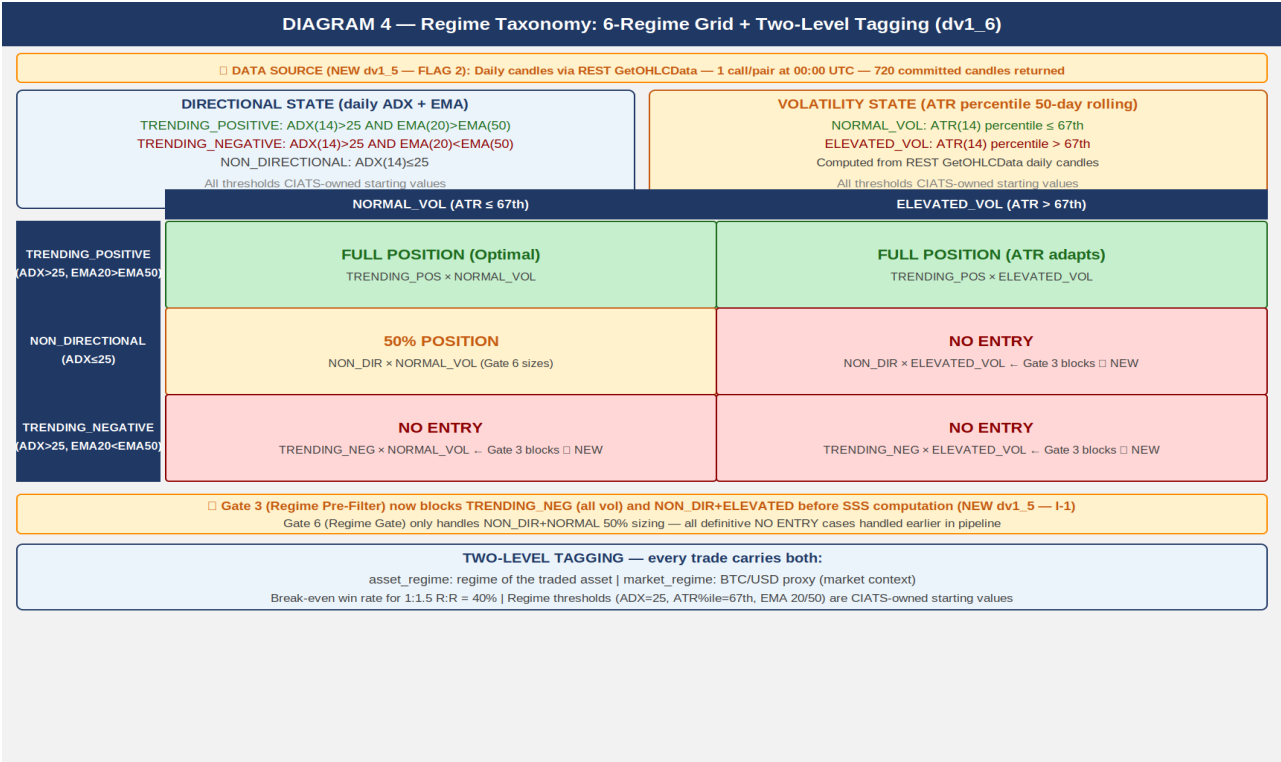
VPS/TothBot/internet failure ONLY. Off-book emergSL stop order at ATR(14) x 3.0x on Kraken matching engine. NEVER in normal operation.
MPP RISK (C-1 dv1_4): triggered market order subject to MPP on wide spread.
Assign to 0411001.

4.2 R:R Formula

mae_pct = ATR(14) x 1.5x / entry_fill_price (CIATS-owned, uses ACTUAL fill price)
net_loss = mae_pct + 0.16% + 0.26% (entry maker + MAE taker exit)
net_gain = net_loss x 1.5 *** SACRED – hardcoded ***
gross_target = net_gain + 0.16% (TP maker exit)
emergSL_dist = ATR(14) x 3.0x / entry_fill_price (NEVER in R:R formula)

NOTE: All calculations use actual fill price, not limit price. OTO uses limit price — this is why OTO was rejected (AR-008).

5. Regime Taxonomy



Six-regime grid is foundation of CIATS regime-segmented analysis. Every trade tagged with both asset_regime and market_regime (BTC proxy). Computed daily at 00:00 UTC.

5.1 Computation

DATA SOURCE (NEW dv1_5 — FLAG 2 — FP/DP VERIFIED):

Daily candles via REST GetOHLCData called once per monitored pair at 00:00 UTC.
Returns 720 entries.

CRITICAL dv1_6 (GAP-03): The last entry (response[-1]) is ALWAYS the current not-yet-committed candle. MUST be excluded from ALL calculations (ADX, EMA, ATR). Use only response[:-1]. Including the uncommitted candle corrupts all regime indicators and produces incorrect regime classification. See 1011012 Regime_Engine_Coding_Spec.

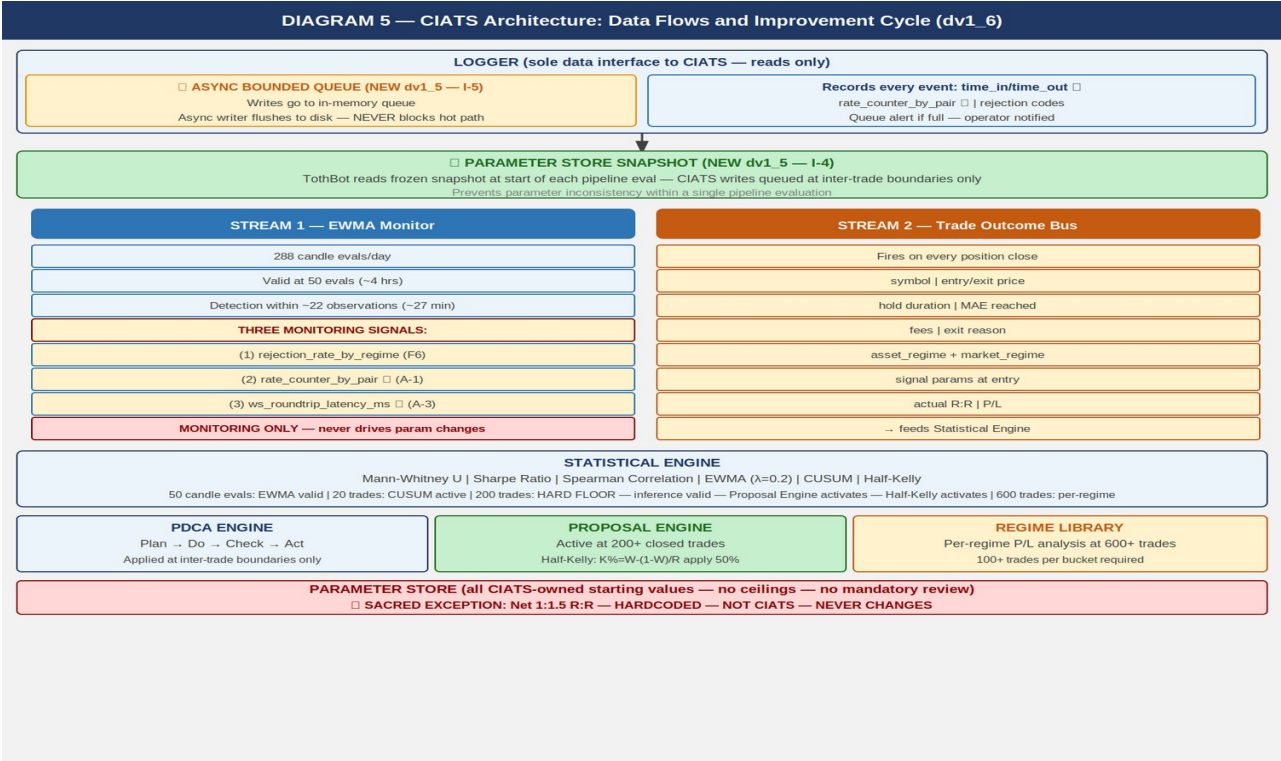
Directional state: ADX(14) + EMA(20) vs EMA(50) on DAILY candles (response[:-1] ONLY)
Volatility state: ATR(14) percentile rank in 50-day rolling window

TRENDING_POSITIVE: ADX(14) > 25 AND EMA(20) > EMA(50)
TRENDING_NEGATIVE: ADX(14) > 25 AND EMA(20) < EMA(50)
NON_DIRECTIONAL: ADX(14) <= 25
NORMAL_VOL: ATR percentile <= 67th
ELEVATED_VOL: ATR percentile > 67th

asset_regime + market_regime (BTC/USD proxy) both tagged on every trade

ALL REGIME THRESHOLDS ARE CIATS-OWNED: ADX threshold 25, ATR percentile 67th, EMA periods 20/50, rolling window 50 days — all starting values. CIATS validates each after 600 closed trades (100+ per regime bucket).

6. CIATS Architecture and Data Flows



CIATS is fully asynchronous. Never blocks the trading critical path. Reads only from the Logger. Writes only to the Parameter Store. Deming PDCA governs every change.

6.1 Two Data Streams

Stream 1: EWMA Monitor

288 candle evals/day. Valid at 50 evals (~4 hrs). $\lambda=0.2$ [FP/DP validated dv1_6: half-life=3.1 evals]. Three monitoring signals:

- (1) rejection_rate_by_regime (F6)
- (2) rate_counter_by_pair from executions channel (A-1)
- (3) ws_roundtrip_latency_ms from WS response timestamps (A-3)

MONITORING SIGNALS ONLY — never drives parameter changes.

Stream 2: Trade Outcome Bus

Fires on each position close. Captures: symbol, entry/exit price, hold duration, MAE reached, fees, exit reason, both regime tags, signal params at entry, actual R:R, P/L.

LOGGER — ASYNC BOUNDED QUEUE (NEW dv1_5 — I-5):

Logger writes go to an in-memory bounded queue. Queue size: 10,000 entries

[FP/DP validated dv1_6: 20 positions x 10 events/candle x 50-candle buffer].

Separate async writer flushes to disk. Logger NEVER blocks the trading hot path.

Queue must be bounded with immediate alert if queue fills. Assign to 0511006.

PARAMETER STORE — SNAPSHOT PATTERN (NEW dv1_5 — I-4):

TothBot takes frozen snapshot of all CIATS parameters at start of each pipeline eval.

Pipeline uses snapshot throughout. CIATS writes queued and applied only at confirmed inter-trade boundaries. Prevents parameter inconsistency within single pipeline eval.

Assign to 0511002 and 0211001.

6.2 Statistical Thresholds

50 candle evals: EWMA Monitor valid. Monitoring only. $\lambda=0.2$ validated.

200 trades HARD FLOOR: Inference valid. Proposal Engine activates. Half-Kelly activates.

SE(win_rate)~3.5%, SE(Sharpe)~7.2% [FP/DP validated dv1_6].

600 trades: Per-regime analysis. 100+ per bucket. MW power ~85% at $\alpha=0.01$.

Rolling 200-300: Current performance window.

Full corpus: Zero-cost historical library.

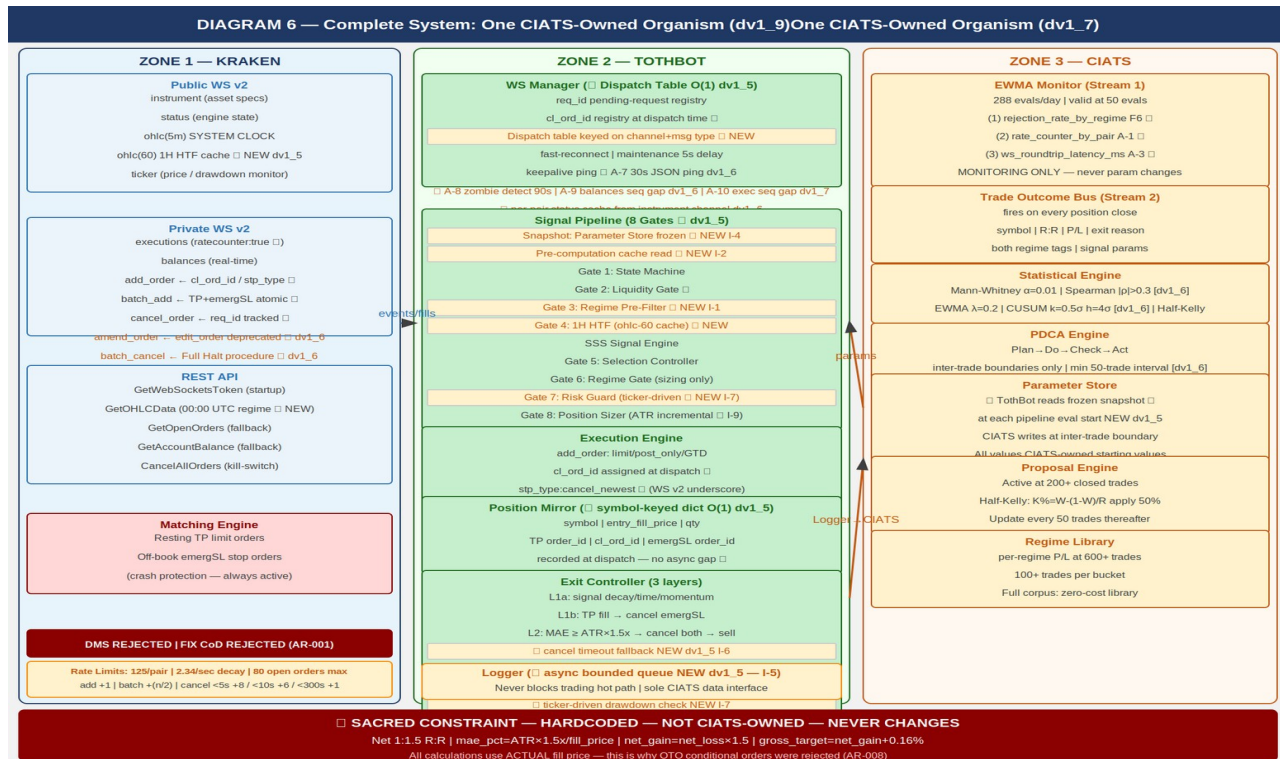
Mann-Whitney alpha (dv1_6) ★: $\alpha=0.01$ (one-sided). 99% confidence required.

Spearman rho threshold (dv1_6) ★: $|\rho| > 0.3$ AND $p < 0.05$ (both conditions).

CUSUM parameters (dv1_6) ★: $k=0.5\sigma$, $h=4\sigma$. Starting values.

PDCA minimum interval (dv1_6) ★: Min 50 trades between any two parameter changes.

7. Complete System: One CIATS-Owned Organism



CIATS OWNS EVERYTHING EXCEPT 1:1.5 R:R: All parameters, thresholds, sizing decisions, halt thresholds, multipliers, and trading rules are CIATS-owned starting values.

8. Locked Architectural Decisions

AR-001: No Dead Man Switch — crash protection is resting TP + emergSL on matching engine.

DMS and FIX cancel-on-disconnect both REJECTED (destroy crash protection).

AR-002: Position Mirror — symbol-keyed dict, O(1) lookup. (NEW dv1_5 — I-3)

AR-003: Restart reconciliation — PREFERRED: snap_orders:true. FALLBACK: REST.

AR-004: Two API key pairs — DATA + TRADE. Nonce window 10,000ms.

AR-005: WS v2 only. All future Kraken enhancements in v2 only.

AR-006: Entry order —

limit/post_only/GTD/cl_ord_id/stp_type:cancel_newest/deadline:now+5s.

★ WS v2 uses underscore in stp_type values (cancel_newest). REST uses hyphen (cancel-newest).

Never confuse the two. Different protocols, different formats.

AR-007: TP placement — resting limit via batch_add on fill. Actual fill price for R:R.
batch_add includes deadline:now+5s and stp_type:cancel_newest on both legs.

AR-008: Emergency SL — stop-market via batch_add atomic with TP. ATR(14) × 3.0x.

OTO evaluated and REJECTED: uses limit price, violates R:R, Position Mirror race.

AR-009: Short deferred.

AR-010: 24/7/365. CIATS adds stand-down via PDCA if data supports.

AR-011: Net 1:1.5 R:R — SACRED — HARDCODED — NOT CIATS-OWNED.

AR-012: Fee baseline — Maker 0.16% / Taker 0.26%.

AR-013: Kraken Pro Intermediate tier confirmed.

AR-014: Logger async bounded queue (10,000 entries). Never blocks hot path. (dv1_5 I-5)

AR-015: WS dispatch table O(1) routing on channel+message type. (dv1_5 I-8)

AR-016: ATR(14) incremental running calculation. O(1) per update. (dv1_5 I-9)

AR-017: Daily regime via REST GetOHLCData at 00:00 UTC. CRITICAL: exclude response[-1]. (dv1_5 FLAG 2)

AR-018 (dv1_6) *: amend_order PRIMARY. edit_order DEPRECATED.

AR-019 (dv1_6) *: batch_cancel for Full Halt multi-order cancellation.

AR-020 (dv1_6) *: Per-pair instrument status cache from instrument channel.

AR-021 (dv1_7) *: Executions channel sequence gap detection → REST GetOpenOrders reconciliation. Missed executions = Position Mirror desync = loss risk.

AR-022 (dv1_8) *: cancel_reason field deprecated – use reason field in all executions channel parsing. cancel_reason returns None silently on current messages.

AR-023 (dv1_8) *: Complete exec_type dispatch table required. All 10 exec_type values handled explicitly. Unknown values logged, never silently dropped.

AR-024 (dv1_8) *: restated exec_type = engine maintenance amend, NOT fill/cancel. Must not trigger Position Mirror update. Elevated alert if resting order.

AR-025 (dv1_8) *: All REST calls via aiohttp (IMP-06). asyncio TaskManager pattern mandatory for 24/7 operation (IMP-07). systemd WatchdogSec=120 with 30-second asyncio ping loop (IMP-08). See 1011013 VPS_Deployment_Spec.

AR-026 (dv1_9) *: order_status:true MANDATORY in executions subscription. Without it amended and restated exec_types are silently dropped – A-12/A-13 blind.

AR-027 (dv1_9) *: interval_begin replaces deprecated timestamp in OHLC WS v2 channel.

AR-028 (dv1_9) *: price_increment replaces deprecated tick_size in instrument channel. qty_min and cost_min validated in Gate 8 before every add_order.

AR-029 (dv1_9) *: WS connect max_size=10MB + open_timeout=10 on all connections.

AR-030 (dv1_9) *: maxratecount from executions subscription ACK is the operative rate counter ceiling. Use it, not the hardcoded 125.

AR-031 (dv1_9) *: Subscription ACK warnings[] parsed and logged on every subscribe call.

QueryOrdersInfo REST endpoint for targeted single-order reconciliation.

AR-032 (dv1_10) *: stp_type FORMAT CORRECTION – WS v2 uses underscores: cancel_newest, cancel_oldest, cancel_both. REST uses hyphens: cancel-newest. Never confuse the two protocols. All WS v2 add_order and batch_add MUST use cancel_newest (underscore). Assign: 0411001, 1011002.

AR-033 (dv1_10) *: deadline parameter on all WS v2 add_order and batch_add. Standard: deadline = UTC now + 5 seconds (ISO 8601 string).

Range 500ms–60s. Engine rejects orders arriving after this time.

Provides latency protection on entry and TP/emergSL placement.

Assign: 0411001 WS-DL-001-004, 1011002 WM-DL-001-004.

AR-034 (dv1_10) *: Ticker event_trigger per position state:

event_trigger:bbo — pairs WITH open positions (faster adverse price detection; for longs, bid = realizable exit price).

event_trigger:trades — pairs WITHOUT open positions (less update volume).

WS Manager switches trigger mode based on Position Mirror state.

Assign: 0411001 WS-TKR-001-004, 1011002 WM-TKR-001-004.

AR-035 (dv1_10) *: aiohttp ClientSession MUST configure explicit timeout + connector: ClientTimeout(total=10, connect=5, sock_read=8)

TCPCConnector(limit=10, limit_per_host=5, force_close=False)

Without timeout, REST calls hang indefinitely on degraded networks.

force_close=False enables connection reuse for reduced per-call overhead.

Assign: 1011002 WM-REST-006-011, 1011001 BP-HTTP-001-004.

AR-036 (dv1_10) ★: Regime Engine REST calls MUST be staggered at 1.1s intervals.
REST public OHLC rate-limit: 1 call/second per IP+pair.
50 pairs × 1.1s = 55s total. Completes before first 5m candle.
asyncio.sleep(1.1) between each GetOHLCData call.

Assign: 1011012 RE-RL-001-004.

AR-037 (dv1_10) ★: Both DATA and TRADE API keys MUST be IP-whitelisted to VPS IP.
Current VPS: 87.99.141.44. Update whitelist on any VPS migration.
IP whitelisting is configured separately in Kraken API key settings.

Assign: 1011013 VD-KEY-006-009.

AR-038 (dv1_12) ★: Kraken Status API used at startup for proactive maintenance detection.

Public REST API at <https://status.kraken.com/api/v2/>. Query
/scheduled-maintenances/upcoming.json on startup. If scheduled

maintenance detected within 2 hours: log
KRAKEN_MAINTENANCE_SCHEDULED,
alert operator. Does NOT block startup — informational only.

Assign: 0411001 WS-STAT-001 through -004, 1011013 VD-STAT-001.

AR-039 (dv1_12) ★: Per-pair instrument channel status check expanded. Complete set of
NO-ENTRY states: reduce_only, work_in_progress, delisted, limit_only,
cancel_only, maintenance. cancel_only and maintenance were previously
missing. Both block all new order placement at the pair level.

Assign: 0411001 WS-INST-008 through -011, 1011002 WM-PS-001 through -
005.

AR-040 (dv1_12) ★: Exit Controller pair status adaptation for open positions.
limit_only → IOC limit at best_bid (NOT market sell).

cancel_only → HOLD + CRITICAL alert. No new orders possible.
maintenance (pair level) → HOLD + CRITICAL alert.

Existing resting TP + emergSL are the only protection during HOLD.

Validates AR-001: resting emergSL crash protection is the correct
architecture for pair-level disruption scenarios.

Assign: 0411001 WS-EXIT-001 through -006, 1011002 WM-PS-003/004.

AR-041 (dv1_12) ★: cancel_order/batch_cancel behavioral distinction — NOT
interchangeable.

cancel_order: returns stream of individual responses per order
(req_id

trackable). Use for Layer 2 MAE exit cancel sequence (cancel timeout
fallback I-6 requires per-order ACK tracking).

batch_cancel: returns single count response + bypasses rate counter
maximum. Use for Full Halt (count sufficient; rate bypass ensures
execution even when rate limited).

Assign: 0411001 WS-CAN-001 through -007, 1011002 WM-CAN-001 through -
004.

AR-042 (dv1_12) ★: Zombie detection heartbeat exclusion. Heartbeat messages fire ONLY
when
no real market data is flowing — they are an indicator of zombie state,
NOT connection health. last_real_data_time MUST be reset only on real
channel events: ohlc, ticker, instrument, executions, balances, status.
Heartbeat, pong, and ACK messages MUST NOT reset the timer.

Assign: 1011002 WM-ZOM-004 through -007.

AR-043 (dv1_12) ★: max_concurrent position limit enforced by asyncio.BoundedSemaphore.
BoundedSemaphore raises ValueError on over-release (bug detection).
Gate 7 checks semaphore.locked() (non-blocking). Execution Engine

acquires semaphore before dispatch. Position close triggers release.
On CIATS max_concurrent update: recreate semaphore, re-acquire for
all currently open positions.

Assign: 1011011 RE-SEM-001 through -006.

9. Complete CIATS Parameter Registry

ALL PARAMETERS ARE CIATS-OWNED STARTING VALUES.

tradeable_pct: 50% of portfolio
per_trade_pct: 5% → Half-Kelly at 200 trades (update every 50 thereafter)

★ Applied_pct = min(raw_kelly_half, tradeable_pct/max_concurrent)

Prevents capital overcommitment across concurrent positions.

See 1011010 CIATS-KE-002 through CIATS-KE-005.

max_concurrent: 20 positions
mae_mult (ATR x): 1.5x ATR(14)
emergency_sl_mult: 3.0x ATR(14)
entry_timeout_sec: 30-60s GTD
full_halt_drawdown: 10% of portfolio
session_pause_drawdown: 5% of portfolio
signal_thresholds: SSS gates (derived)
adx_threshold: 25
atr_percentile_threshold: 67th
htf_ema_periods: 20/50 daily
min_volume_usd_daily: \$500k USD
1:1.5 R:R net: HARDCODED — SACRED — NOT CIATS-OWNED

10. Six FP/DP Validated System Improvements

Derived from 80+ sources across 3 independent research passes plus 2 deep validation
sessions (144+ sources) plus 10-search Kraken validation [dv1_6] (126 sources).
5+ sources per parameter.

Finding 1: 1H HTF Confirmation Gate [TothBot]

Gap: No layer between daily regime and 5-min entry.
Evidence: 15-20% higher win rates with daily+hourly filter (5 sources 2024-2025).
Change: 1H EMA(20) > EMA(50) required for TRENDING_POSITIVE. Cache from ohlc(60).
Effect: Removes false signals during 1H pullbacks. Zero API cost.

Finding 2: ATR Dynamic Stop Distances [TothBot/Exit]

Gap: Fixed 2% stop ignores real-time market volatility.
Evidence: +15% performance, -22% max drawdown vs fixed stops (LuxAlgo + 4 sources).
Change: MAE ATR(14) x 1.5x. Emergency SL ATR(14) x 3.0x. Incremental (I-9).
Effect: Stop adapts to conditions. R:R formula preserved.

Finding 3: Regime Gate [TothBot]

Gap: Long momentum system trading in absent-edge conditions.

Evidence: Accuracy 65%→45-50% in NON_DIRECTIONAL (Market Cipher). Break-even 40%.

Change: NON_DIR+ELEV and TRENDING_NEG: no entry (Gate 3). NON_DIR+NORM: 50% (Gate 6).

Daily candles via REST GetOHLCData. CRITICAL: exclude response[-1] [dv1_6].

Effect: Stops trading when documented edge is absent. Pure loss minimization.

Finding 4: Minimum Liquidity Gate [TothBot]

Gap: No minimum 24h volume on asset selection.

Evidence: \$1M+/day minimum for clean execution (Coin Bureau + 4 sources 2024-2025).

Change: min_24h_volume_usd \$500k starting. Checked daily via ticker.

Effect: Protects fee formula. Ensures clean post_only fills and SL execution.

Finding 5: CIATS Half-Kelly Pathway [CIATS]

Gap: Fixed 5% sizing had no defined data-driven transition.

Evidence: Half-Kelly: 75% optimal growth at 50% less drawdown (MacLean-Thorp-Ziemba + 4).

FP/DP validated [dv1_6].

Change: At 200 trades: $K=W-(1-W)/R$, apply 50%. Update every 50 trades. No ceilings.

Effect: Data-driven sizing replaces arbitrary fixed fraction.

Finding 6: post_only Rejection Monitoring [CIATS]

Gap: Rejection events logged but unused by CIATS EWMA.

Evidence: Rejection rate = direct signal of entry placement quality (Cube Exchange + 4).

Change: CIATS EWMA adds rejection_rate_by_regime stream.

Effect: Free diagnostic signal. Entry optimization pathway activated.

11. Complete Trade Lifecycle — Narrative Reference

11.1 Normal Trade (Entry to TP or Exit Controller)

OHLC 5-min candle close fires on public WS v2

Parameter Store snapshot frozen — all CIATS params captured (NEW dv1_5 — I-4)

Pre-computation cache read — regime, 1H EMA, ATR(14) incremental, 24h liquidity (I-2)

★ Per-pair instrument status check — skip reduce_only/work_in_progress/delisted/limit_only/cancel_only/maintenance [dv1_12]

Gate 1: State Machine — online or post_only: continue; else halt and wait

Gate 2: Liquidity Gate — 24h volume $\geq$ \$500k: continue; else skip

Gate 3: Regime Pre-Filter — TRENDING_NEG or NON_DIR+ELEV: STOP; else continue (I-1)

Gate 4: 1H HTF Gate — 1H EMA(20) > EMA(50) for TRENDING_POS: continue; else skip

Signal Engine: SSS score computed on 5-min candle

Gate 5: Selection Controller — 6-gate quality evaluation: pass or reject

Gate 6: Regime Gate — TRENDING_POS: proceed; NON_DIR+NORM: 50% size

Gate 7: Risk Guard — drawdown/concentration/exposure checks: pass or halt

Gate 8: Position Sizer — $MAE=ATR \times 1.5x$, $emergSL=ATR \times 3.0x$, $TP=R:R$ formula

Execution: WS add_order (limit/post_only/GTD

30-60s/cl_ord_id/stp_type:cancel_newest/deadline:now+5s)

If expired/rejected: log reason code → CIATS EWMA monitors rejection rate. Done.
If filled: WS batch_add → TP (resting limit) + emergSL (stop-market) atomic.
cl_ord_id/stp_type:cancel_newest/deadline:now+5s
Position Mirror records: symbol (key), entry_fill_price, qty, TP id, cl_ord_id, emergSL id
Exit Controller monitors via ticker (continuous drawdown check — I-7)
Logger records full entry via async queue. CIATS Trade Outcome Bus ready.
TP fills: EC cancels emergSL. Or EC fires Layer 1/2 exit.
Logger records full exit. CIATS Trade Outcome Bus fires with complete trade record.

11.2 Layer 2 MAE Exit (Adverse Price)

Exit Controller detects: $\text{current_price} - \text{entry} \geq \text{ATR}(14) \times 1.5x$
WS cancel_order: cancel TP resting order
WS cancel_order: cancel emergSL stop order (SEQUENCE CRITICAL: both cancels before sell)

CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6):
req_id registry tracks each cancel ACK. On timeout: check executions channel state.
Confirmed cancelled: proceed. State unknown: retry once. Second timeout: ALERT and hold position. NEVER issue market sell with ambiguous order state.
Assign to 0511001 and 0511004.

WS add_order: market sell (taker 0.26%)
MPP NOTE (C-1): Market sell subject to MPP. EC MUST handle MPP rejection and retry.
Logger: exit reason = MAE_THRESHOLD_BREACH
CIATS Trade Outcome Bus fires

11.3 Emergency SL (Layer 3 — System Failure Only)

VPS crashes / TothBot dies / internet severs / unrecoverable WS disconnect
Kraken matching engine fires off-book emergSL stop-market at $\text{ATR} \times 3.0x$
MPP NOTE: Triggered market order subject to MPP. Assign to 0411001.
On restart: reconciliation detects closed position → Logger: EMERGENCY_SL_FIRED
THIS IS A SYSTEM FAILURE EVENT. SHOULD NEVER FIRE IN NORMAL OPERATION.

12. Open Items — Require Specification Documents

OI-001: Signal generation (SSS) — FP/DP derivation required. Blocks 0811001, 0811002.
OI-002: Asset selection universe — Kraken USDT/USDC pairs.
OI-003: Exit Controller full specification — MPP handling, cancel timeout fallback, ticker-driven drawdown, batch_cancel for Full Halt [dv1_6].
OI-004: Entry limit placement — evaluate bbo trigger, L2 book (PL-010), L3 book wss://ws-l3.kraken.com/v2 [URL corrected dv1_6].
OI-005: Selection Controller — 6-gate spec, must include per-pair status check [dv1_6].
OI-006: CIATS specification — Mann-Whitney $\alpha=0.01$, Spearman $|\rho|>0.3$, CUSUM k/h,

PDCA 50-trade interval [dv1_6].

OI-007: AA=04 Exchange Interface formal document.

OI-008: All AA=10 coding specs. WD documents active: 1011001, 1011002, 1011010, 1011012,

1011013. All dv1_2 documents: 0411001 dv1_2, 1011002 dv1_2 (updated dv1_9).

13. Revision History

dv1_0 03-23-2026 Initial document.
dv1_1 03-23-2026 Added all six system flow diagrams.
dv1_2 03-24-2026 Six changes: C-1 emergSL terminology, C-2 rate limits, C-3 OHLC stall,

A-1 ratecounter, A-2 req_id, A-3 latency logging.

dv1_3 03-24-2026 Rebuilt as correct .docx with all six diagrams embedded.

dv1_4 03-24-2026 Seven changes: C-1 MPP, A-1 AR-001 expansion, A-2 cl_ord_id, A-3 snap_orders, A-4 stp_type, A-5 reconnect distinction, A-6 OT0 rejected.

dv1_5 03-24-2026 Eleven changes: FLAG1 ohlc(60), FLAG2 REST GetOHLCData, I-1 Gate 3, I-2 cache, I-3 Position Mirror O(1), I-4 Parameter Store snapshot, I-5 Logger async queue, I-6 cancel timeout, I-7 ticker drawdown, I-8 WS dispatch table, I-9 ATR incremental.

dv1_6 03-24-2026 Six changes from 10-search FP/DP validation + Operations/CIATS analysis:

(1) amend_order primary, edit_order deprecated, batch_cancel, L3 URL corrected.

(2) Rate table updated.

(3) A-7 WS ping, A-8 zombie detection, A-9 balances seq gap.

(4) Per-pair instrument status check, AR-018/019/020.

(5) GetOHLCData response[-1] exclusion — CRITICAL correctness requirement.

(6) Auth verification, CIATS statistical thresholds.

dv1_9 03-25-2026 Six new architectural requirements and five improvements from second 10-search FP/DP organism validation session (TB00010):

A-15 (GAP-A CRITICAL) order_status:true mandatory in executions subscription — without it amended/restated silently dropped, breaking

A-12/A-13. A-16 (GAP-B) OHLC interval_begin replaces deprecated timestamp. A-17 (GAP-C/D) instrument price_increment replaces

tick_size; qty_min/cost_min mandatory Gate 8 validation. A-18 (GAP-E)

websockets max_size=10MB + open_timeout=10 mandatory. A-19 (IMP-11)

maxratecount from executions ACK is operative rate threshold. IMP-13

subscription warnings[] parsed/logged. IMP-14 amend_id logged. IMP-15

QueryOrdersInfo for targeted reconciliation. AR-026 through AR-031 added.

0411001 updated to dv1_2. 1011002 updated to dv1_2.

dv1_10 03-25-2026 Seven new findings from third 10-search FP/DP organism validation (TB00011): A-20 stp_type FORMAT CORRECTION. A-21 deadline parameter.

A-22 aiohttp ClientTimeout + TCPConnector. A-23 Ticker event_trigger:bbo.

A-24 Regime Engine GetOHLCData stagger. AR-037 IP-whitelisting.

IMP-16 through IMP-19. AR-032 through AR-037. 0411001 dv1_3.
1011002 dv1_3. 1011012 dv1_1. 1011001 dv1_1.

dv1_12 03-25-2026 Ten text-only changes from fourth (TB00012), fifth (TB00013), and sixth (TB00014) 10-search FP/DP validation sessions. No diagram changes.

AR-038: Kraken Status API startup maintenance check. Section 2 REST table: Status API endpoint added. AR-039: per-pair status check expanded — cancel_only and maintenance added as NO-ENTRY states.

AR-040: Exit Controller pair status adaptation — IOC limit on limit_only,
HOLD on cancel_only/maintenance. AR-041: cancel_order/batch_cancel behavioral distinction — cancel_order for MAE exit (individual ACK),
batch_cancel for Full Halt (count + rate bypass). AR-042: zombie detection heartbeat exclusion — heartbeats MUST NOT reset last_real_data_time. AR-043: max_concurrent via asyncio.BoundedSemaphore.
Section 9 CIATS Parameter Registry: Half-Kelly normalization note.
Section 11.1: per-pair status check updated with full list.
0411001 updated to dv1_6. 1011002 updated to dv1_6.
1011010 updated to dv1_2. 1011011 dv1_0 new.
1011012 updated to dv1_2. 1001001 updated to dv1_3.

dv1_8 03-25-2026 Eight changes from 10-search FP/DP organism validation session (TB00009 Part 2): A-11 executions cancel_reason deprecated — use reason field. A-12 complete exec_type enum handling required. A-13 restated exec_type handling — engine amend, not fill/cancel. A-14 PL-014 RESOLVED — Spot REST auth confirmed unchanged (Oct 2025 change was Futures REST only). IMP-06 aiohttp mandatory for all REST calls.

IMP-07 asyncio TaskManager pattern for 24/7 memory safety. IMP-08 systemd WatchdogSec=120 with 30s ping loop. New AR-022 through AR-025.
New WD document 1011013 VPS_Deployment_Coding_Spec dv1_0 created.
0411001 updated to dv1_1. 1011002 updated to dv1_1.
Diagrams 1 and 6 updated.

dv1_7 03-25-2026 One change from 10-search FP/DP organism validation session (TB00009):

A-10: Executions channel sequence gap detection. Every executions message carries a sequence integer. WS Manager tracks it; any gap triggers REST GetOpenOrders reconciliation immediately. Missed execution events leave Position Mirror in stale state — highest-severity gap class. AR-021 added. GetOpenOrders promoted from restart-only fallback to active gap recovery channel. Diagrams 1 and 6 updated with A-10.